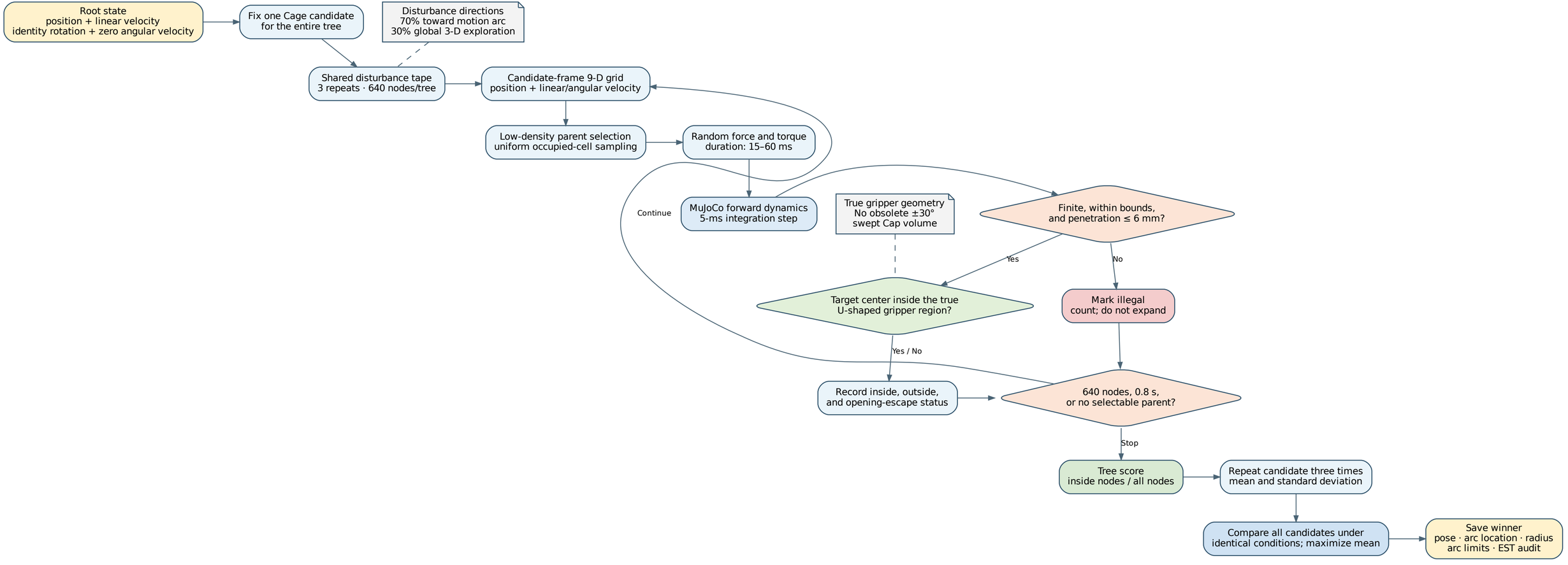
Root state
position + linear velocity
identity rotation + zero angular velocity
Fix one Cage candidate
for the entire tree
Disturbance directions
70% toward motion arc
30% global 3-D exploration
Shared disturbance tape
3 repeats · 640 nodes/tree
Candidate-frame 9-D grid
position + linear/angular velocity
Low-density parent selection
uniform occupied-cell sampling
Random force and torque
duration: 15–60 ms
MuJoCo forward dynamics
5-ms integration step
Continue
True gripper geometry
No obsolete ±30°
swept Cap volume
Finite, within bounds,
and penetration ≤ 6 mm?
Yes
No
Target center inside the true
U-shaped gripper region?
Mark illegal
count; do not expand
Yes / No
Record inside, outside,
and opening-escape status
640 nodes, 0.8 s,
or no selectable parent?
Stop
Tree score
inside nodes / all nodes
Repeat candidate three times
mean and standard deviation
Compare all candidates under
identical conditions; maximize mean
Save winner
pose · arc location · radius
arc limits · EST audit